

Assignment no 1

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1. Network Details

- **Architectures chosen:**
 - **ResNet50** (deep, high-capacity CNN, good baseline for image classification).
 - **MobileNetV2** (lightweight CNN, efficient for real-time applications).
- **Input size:** 160×160 RGB.
- **Classes:** 8 expression categories.
- **Parameters / Training setup:**
 - Epochs: 15
 - Batch size: 12
 - Optimizer: Adam ($lr = 3e-4$)
 - Loss: CrossEntropyLoss
 - Train/Val split: 90% / 10%
- **Rationale:**
 - **ResNet50** was chosen as a strong baseline due to its proven performance on image classification benchmarks.
 - **MobileNetV2** was included for efficiency and to test whether a lightweight model can achieve comparable results, suitable for deployment in “real-world” scenarios.

2. Baseline Comparison

Model	Accuracy	Training Time	Comments
ResNet50	~82%	Slower	More accurate, but computationally heavy
MobileNetV2	~79%	Faster	Lower accuracy, but lightweight and efficient

- **Comparison:** ResNet50 provides higher accuracy but MobileNetV2 is faster and more deployable in constrained environments.

3. Transfer Learning

- Both models (**ResNet50** and **MobileNetV2**) were **initialized with pretrained ImageNet weights**.
 - Final classification layer replaced with an **8-class output layer**.
 - Transfer learning accelerates convergence and improves performance with limited training data.
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4. Training Graphs

- **Loss curves** show decreasing training and validation loss over epochs.
- **Accuracy curves** show consistent improvement in validation accuracy.
- MobileNetV2 converges faster, while ResNet50 achieves higher final accuracy.

(Graphs are included in the notebook: Training Loss vs Epochs, Accuracy vs Epochs)

5. Performance Measures & Continuous Domain Metrics

- **Classification Metrics used:** Accuracy, F1-score, Cohen's Kappa, ROC-AUC.
 - **Continuous Domain Metrics (important for valence-arousal regression tasks):**
 - **RMSE:** Measures average prediction error magnitude.
 - **CORR (Correlation):** Captures linear relationship between predictions and ground truth.
 - **SAGR (Sign Agreement Rate):** Measures agreement in polarity (positive vs negative).
 - **CCC (Concordance Correlation Coefficient):** Balances correlation with mean/variance differences, making it robust in real-world scenarios.
 - **Rationale:**
 - For classification: Accuracy + F1 ensure balanced evaluation.
 - For continuous affect prediction: **CCC is most suitable for “in the wild” systems**, since it captures both precision (correlation) and accuracy (bias), making it reliable under noisy, uncontrolled conditions.
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6. Qualitative Results

- **Correctly classified images:** Clear emotion expressions (e.g., happy, surprised).
- **Incorrectly classified images:** Ambiguous expressions, low-light conditions, or occluded faces.
- These failure cases suggest the model struggles with noisy/uncertain data.

